CogniLit AI

The Research Assistant

"Read Less, Discover More"



Team Name: HackSmiths

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Selected Domain: D3-PS4 Research Assistant AI Agent

KEY PROJECT DETAILS

PROBLEM

- Academic literature review is a Research Bottleneck: a slow, inefficient process that fails to scale with data volume, forcing time on filtering noise over generating insight.
- The bottleneck stems from High Cognitive Load (information overload). This causes Wasted Effort on manual filtration due to a high Noise-to-Signal Ratio. The consequence is a Productivity Block, delaying critical analysis.

SOLUTION

CogniLit AI is an intelligent agent designed to autonomously address this research bottleneck.

CORE TECHNOLOGY

The platform uses the Gemini API and academic indexes (OpenAlex, ArXiv) for sourcing and analysis

KEY FUNCTIONALITY

It finds high-quality papers, summarizes their key contributions and methods, and ranks them by relevance.

IMPACT

Drastically cuts time to research insight by automating filtration and synthesis. Focus on original work.

PROBLEM STATEMENT

Researchers face difficulty finding relevant resources and training opportunities amid vast online content, hindering effective learning, skill enhancement, and research growth.

01



Literature Overload

Researchers face overwhelming literature volume, wasting time and slowing down discovery of relevant academic work.

02



Delayed Understanding

Extracting core contributions from dense papers is slow and mentally taxing, hindering quick knowledge acquisition.

03



Prioritization Confusion

It's hard to assess true paper quality or relevance, leading to misprioritized effort and wasted reading time.

04



Inefficient Exploration

Starting new research is inefficient and friction-filled, delaying the critical move from query to focused analysis.

PROPOSED SOLUTION

CogniLit AI

The Research Assistant

Go from question to ranked, citable insight instantly. Shift your focus from tedious search logistics to high-value analysis.

Autonomous Research Agent

The agent autonomously queries specialized academic sources to filter noise and synthesize findings into structured data(methods, rankings, etc.).

Conversational & Contextual History

Maintains full session memory and search context, allowing users to seamlessly build upon past insights and refine research iteratively.

Personalized Research Workspace

Secure user accounts with persistent storage for saved artifacts, chat history, and personalized settings. Offers secure access via both manual and social login.

Optimized, Citable Output

Delivers highly relevant results ranked by importance and grounded with verifiable source links, enabling the user to jump directly to core insights.

INNOVATION AND UNIQUENESS

CogniLit AI is not a search engine; it's a Synthesis Engine. It fundamentally changes the user's workflow from manual filtration to automated analysis, cutting the research time and dramatically increasing productivity.

Persistent Context

Transforms literature into analyzable JSON and memory, accelerating research insight.



Insight Synthesis

The agent moves beyond vast PDF lists by autonomously synthesizing and ranking insights from multiple sources, providing a single, citable answer.



Data Structure

We utilize LLM schemas to enforce strict JSON output, transforming raw paper text into structured, instantly-analyzable data (methods, rankings, URL).



Persistent Memory

The platform maintains full session context via PostgreSQL, enabling seamless query refinement and iterative research within a secure workspace.



Academic Grounding

Intelligence relies only on academic indexes (ArXiv, OpenAlex), minimizing general web noise and guaranteeing verifiable, citable academic integrity.

TECHNICAL FEASIBILITY

Dedicated AI Agent Core

1

The Gemini + LangChain core is isolated into a service, ensuring research complexity never burdens the core API or UI performance.

Modular Backend Pattern

2

Flask Blueprints facilitate rapid, parallel feature development, enabling clean code separation and seamless future integration.

Structured Output Enforcement

3

Guarantees immediate, consumable results by enforcing structured JSON output, eliminating frontend parsing errors.



Optimized Dual Persistence

6

Balances data integrity (PostgreSQL) with zero-latency performance (Redis for ephemeral data), optimizing both safety and speed.

Secure Client-Side Auth Flow

5

React manages the OAuth token flow and validation, while Flask securely creates sessions, ensuring modern security compliance.

Optimized UI Synchronization

4

A single /auth/status API call on load minimizes network overhead, instantly synchronizing the UI state with the server.